

BROOKS, AB, Canada

WMO: 714570

Lat: 50.560N

Lon: 111.850W

Elev: 2451

StdP: 13.44

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-16.2	-29.0	1.2	-21.4	-22.7	1.7	-16.0	26.6	24.9	23.8	23.4	4.0	180	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.8	89.0	63.6	85.5	62.7	82.1	61.6	67.2	81.6	65.4	79.7	63.6	77.8	9.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	93.3	72.3	60.5	86.1	70.0	58.4	80.0	68.4	33.1	82.1	31.6	79.4	30.2	78.0	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	20.4	17.8	-30.2	95.0	7.1	3.2	-35.3	97.2	-39.5	99.1	-43.5	100.9	-48.6	103.2		
(5)			-30.4	71.8	7.0	4.1	-35.5	74.7	-39.6	77.1	-43.6	79.4	-48.7	82.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	14.6	19.0	29.0	41.8	52.3	60.0	66.4	64.4	55.0	42.1	27.7	16.5	(6)
(7)		DBStd	21.31	16.21	14.35	14.56	8.78	7.87	5.83	5.46	5.81	7.82	9.08	13.47	15.01	(7)
(8)		HDD50	4976	1099	868	653	266	66	3	0	1	38	271	673	1038	(8)
(9)		HDD65	9006	1564	1288	1116	696	397	171	48	83	310	710	1120	1503	(9)
(10)		CDD50	1634	0	0	1	20	139	303	508	446	188	26	3	0	(10)
(11)		CDD65	190	0	0	0	0	4	21	92	63	10	0	0	0	(11)
(12)		CDH74	3515	0	0	1	15	194	393	1427	1134	339	12	0	0	(12)
(13)		CDH80	1165	0	0	0	1	42	90	522	409	101	0	0	0	(13)
(14)	Wind	WSAvg	7.9	7.5	7.3	8.4	9.3	9.1	8.3	7.5	7.1	7.3	7.9	7.6	7.3	(14)
(15)	Precipitation	PrecAvg	13.0	0.6	0.4	0.6	1.0	1.7	2.6	1.6	1.5	1.3	0.6	0.7	0.6	(15)
(16)		PrecMax	21.8	1.1	0.9	1.7	3.1	3.5	7.7	4.9	3.4	3.9	1.7	1.8	1.0	(16)
(17)		PrecMin	6.6	0.1	0.0	0.0	0.1	0.3	0.3	0.1	0.0	0.1	0.2	0.1	0.0	(17)
(18)		PrecStd	3.0	0.3	0.2	0.4	0.6	0.9	1.6	1.3	0.7	0.9	0.4	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	53.7	64.6	75.8	85.0	87.6	93.7	92.9	88.3	75.2	63.4	52.2	(19)
(20)			MCWB	40.8	42.0	47.5	53.2	58.4	63.9	66.5	63.1	60.1	55.2	48.7	40.8	(20)
(21)		2%	DB	44.4	46.5	58.5	69.8	79.5	82.3	89.6	88.6	82.7	68.3	56.3	46.2	(21)
(22)			MCWB	38.1	37.8	44.7	50.4	57.0	61.4	65.1	62.7	58.8	52.0	44.7	38.3	(22)
(23)		5%	DB	41.1	42.0	53.5	64.3	74.6	78.5	85.8	84.8	77.5	63.2	51.1	42.0	(23)
(24)			MCWB	36.2	35.8	42.0	47.4	54.9	59.7	64.2	62.5	57.3	49.5	41.9	36.0	(24)
(25)		10%	DB	37.6	38.2	48.0	59.2	69.9	74.9	82.1	80.7	72.3	58.4	46.0	37.9	(25)
(26)			MCWB	33.8	33.5	39.5	44.9	52.7	58.1	63.0	61.7	55.3	46.8	38.6	33.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.9	42.5	48.6	54.7	62.1	67.9	71.6	68.6	63.4	56.9	49.3	42.8	(27)
(28)			MCD B	47.9	52.0	61.9	72.2	77.3	80.9	85.7	82.1	82.3	70.4	61.2	50.4	(28)
(29)		2%	WB	38.6	38.6	45.7	51.2	58.6	64.0	68.1	66.3	60.5	53.3	45.6	39.0	(29)
(30)			MCD B	43.5	45.3	56.7	67.3	74.4	77.7	83.2	80.3	77.4	65.9	55.6	45.1	(30)
(31)		5%	WB	36.4	36.0	42.6	48.4	56.3	61.9	66.2	64.5	58.2	50.3	42.3	36.2	(31)
(32)			MCD B	40.7	41.5	52.1	62.2	71.5	74.6	80.9	79.1	74.0	61.8	50.5	41.7	(32)
(33)		10%	WB	33.7	33.6	39.8	45.7	54.0	59.8	64.5	62.8	56.1	47.5	39.1	33.4	(33)
(34)			MCD B	37.8	38.0	47.5	57.4	67.8	71.5	78.7	77.2	70.5	57.3	45.2	38.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.3	21.6	22.4	25.3	25.8	23.7	26.8	27.5	27.1	25.1	21.2	20.1	(35)
(36)			MCDBR	22.8	25.8	30.3	36.3	34.5	30.4	33.4	35.8	38.1	32.9	27.2	23.0	(36)
(37)			MCWBR	17.3	18.0	17.9	18.7	15.7	13.2	13.3	14.5	16.9	17.8	17.0	16.2	(37)
(38)		5% WB	MCDBR	22.2	25.1	28.8	33.4	30.5	27.5	29.7	30.6	34.3	31.6	26.5	22.8	(38)
(39)			MCWBR	17.2	18.2	18.0	18.1	14.4	13.5	13.3	13.8	16.1	17.8	17.3	16.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.272	0.283	0.302	0.351	0.374	0.365	0.363	0.385	0.336	0.296	0.268	0.257	(40)
(41)		taud		2.386	2.426	2.436	2.299	2.283	2.364	2.392	2.303	2.424	2.511	2.479	2.395	(41)
(42)		Ebn at Noon		246	274	288	282	279	283	281	268	272	266	248	235	(42)
(43)		Edh at Noon		19	24	29	37	40	37	36	37	29	22	17	16	(43)
(44)	All-Sky Solar Radiation	RadAvg		378	691	1123	1553	1850	1990	2161	1741	1246	764	418	289	(44)
(45)		RadStd		36	63	79	110	125	121	104	112	108	84	36	26	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page